

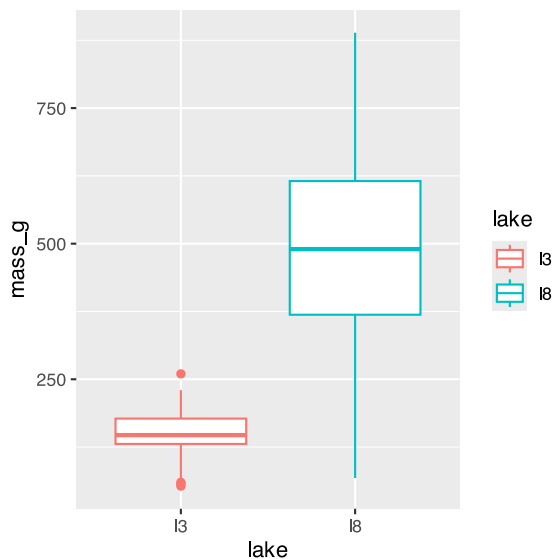
Lecture 09 Correlation and Regression

Bill Perry

Lecture 8: Review

Covered

- Study design
- Causality in ecology
- Experimental design:
 - Replication, controls, randomization, independence
- Sampling in field studies
- Power analysis: *a priori* and *post hoc*
- Study design and analysis



Lecture 9: Overview

The objectives:

This lecture covers two fundamental statistical techniques in biology: correlation and regression analysis. Based on Chapters 16-17 from Whitlock & Schluter's *The Analysis of Biological Data* (3rd edition), we'll explore:

- Correlation analysis: measuring relationships between variables
- The distinction between correlation and regression
- Simple linear regression: predicting one variable from another
- Estimating and interpreting regression parameters
- Testing assumptions and handling violations
- Analysis of variance in regression
- Model selection and comparison

Lecture 9: Correlation vs. Regression:

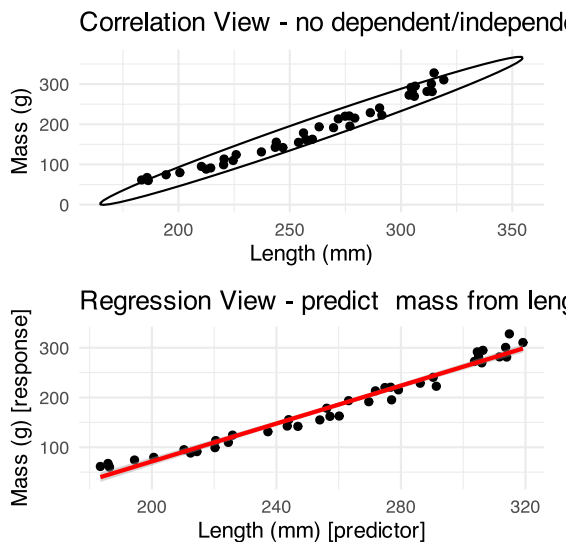
What's the Difference?

Correlation Analysis:

- Measures the strength and direction of a relationship between two numerical variables
- Both X and Y are random variables (both measured, neither manipulated)
- Variables are typically on equal footing (either could be X or Y)
- No cause-effect relationship implied
- Quantifies the degree to which variables are related
- Expressed as a correlation coefficient (r) from -1 to +1

Regression Analysis:

- Predicts one variable (Y) from another (X)
- X is often fixed or controlled (manipulated)
- Y is the response variable of interest
- Often implies a cause-effect relationship
- Produces an equation for prediction
- Estimates slope and intercept parameters



Lecture 9: Correlation Analysis

What Is Correlation?

Correlation analysis measures the strength and direction of a relationship between two numerical variables:

- Ranges from -1 to +1
- +1 indicates perfect positive correlation
- 0 indicates no correlation
- -1 indicates perfect negative correlation

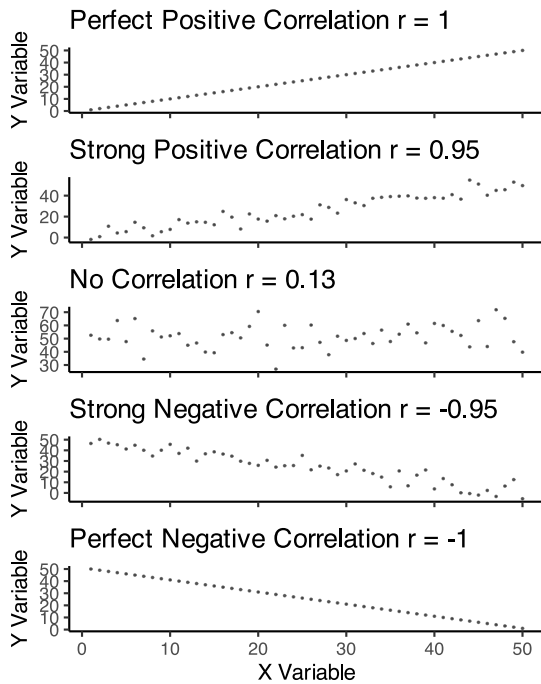
The **Pearson correlation coefficient (r)** is defined as:

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}$$

This can be simplified as:

$$r = \frac{\text{Covariance}(X, Y)}{s_X \cdot s_Y}$$

Where s_X and s_Y are the standard deviations of X and Y.



Lecture 9: Correlation Analysis

Example 16.1: Flipping the Bird

Nazca boobies (*Sula granti*) - Do aggressive behaviors as a chick predict future aggressive behavior as an adult?

- correlation is $r = 0.534$ - moderate positive relationship
- p-value = 0.007 correlation is statistically significant.

For a Pearson correlation coefficient (r) of 0.53372:

- This is r (not rho as Spearman nonparametric below), as indicated by “cor” in your output
- To determine the amount of variation explained, you square this value: $r^2 = 0.53372^2 = 0.2849$ (or approximately 28.49%)
- means about 28.49% of the variance in one variable can be explained by the other variable

Note that it is tested by: $t = \frac{r}{SE_r}$

```
[1] 0.5337225
```

Pearson's product-moment correlation

```
data: booby_data$visits_as_nestling and booby_data$future_aggression
t = 2.9603, df = 22, p-value = 0.007229
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1660840 0.7710999
sample estimates:
```

```
cor
0.5337225
```

Lecture 9: Correlation Analysis

Example 16.1: Flipping the Bird

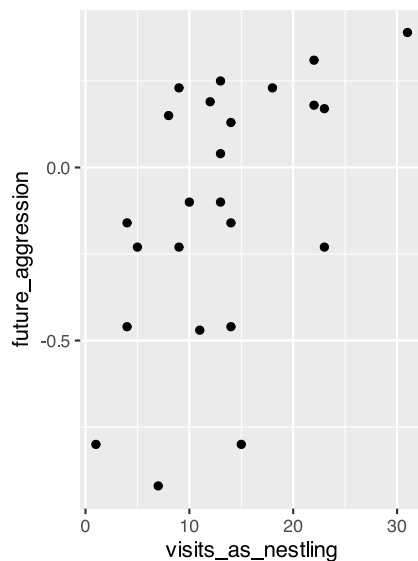
Interpretation: The correlation coefficient of $r = 0.534$ suggests that Nazca boobies who experienced more visits from non-parent adults as nestlings tend to display more aggressive behavior as adults. This supports the hypothesis that early experiences influence adult behavior patterns in this species.

Standard Error:

$$SE_r = \sqrt{\frac{1-r^2}{n-2}}$$

SE = 0.180

Need to be sure relationship is not curved - note below



Lecture 9: Correlation Analysis

Testing Assumptions for Correlation

As described in Section 16.3, correlation analysis has key assumptions:

1. **Random sampling:** Observations should be a random sample from the population
2. **Bivariate normality:** Both variables follow a normal distribution, and their joint distribution is bivariate normal
3. **Linear relationship:** The relationship between variables is linear, not curved

Let's check these assumptions booby data:

Shapiro-Wilk normality test

```
data: booby_data$visits_as_nestling
W = 0.95783, p-value = 0.3965
```

Shapiro-Wilk normality test

```
data: booby_data$future_aggression
W = 0.91575, p-value = 0.04709
```

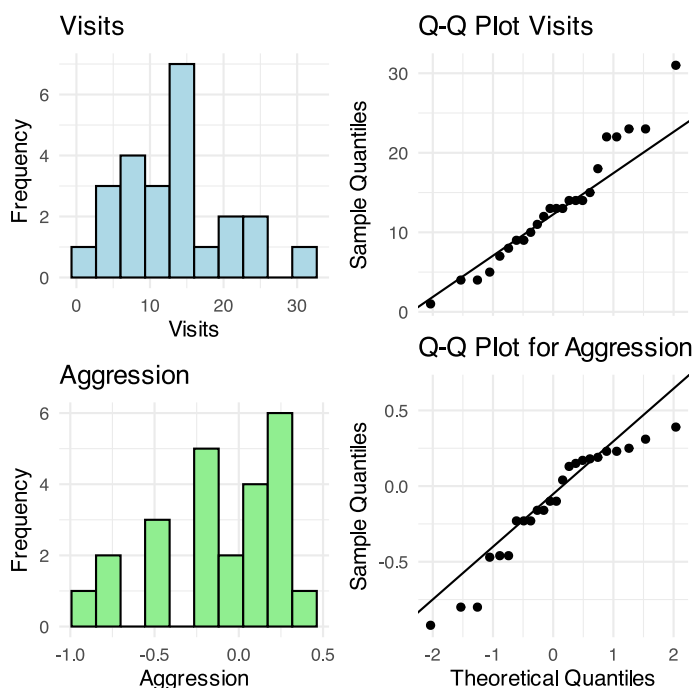
Lecture 9: Correlation Analysis

Testing Assumptions for Correlation

As described in Section 16.3, correlation analysis has key assumptions:

1. **Random sampling:** Observations should be a random sample from the population
2. **Bivariate normality:** Both variables follow a normal distribution, and their joint distribution is bivariate normal
3. **Linear relationship:** The relationship between variables is linear, not curved

Let's check these assumptions using the booby data



Lecture 9: Correlation Analysis

What to do if assumptions are violated:

Transform one or both variables (log, square root, etc.)

Use non-parametric correlation (**Spearman's rank correlation**) or Kendall's tau [?](#)

Examine the data for outliers or influential points

```
#| echo: false
#| fig-height: 5
#| fig-width: 5
#| message: false
#| warning: false

p1_lion <- ggplot(lion_data, aes(x = proportion_black)) +
```

```

geom_histogram(bins = 10, fill = "lightblue", color = "black") +
labs(title = "Prop black",
      x = "Visits",
      y = "Frequency") +
theme_minimal()

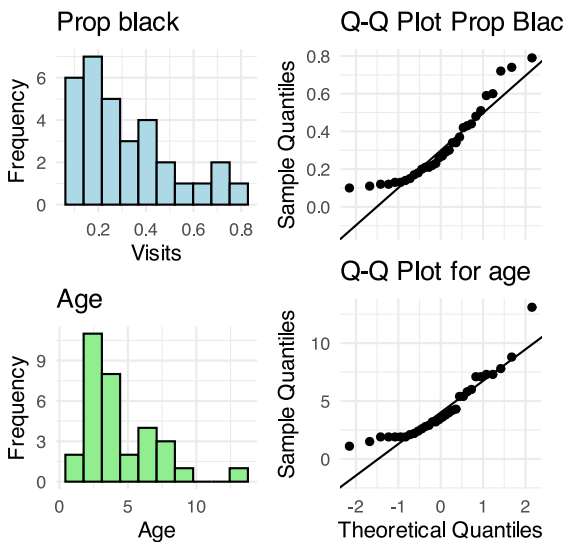
p2_lion <- ggplot(lion_data, aes(x = age_years)) +
  geom_histogram(bins = 10, fill = "lightgreen", color = "black") +
  labs(title = "Age",
        x = "Age",
        y = "Frequency") +
  theme_minimal()

p3_lion <- ggplot(lion_data, aes(sample = proportion_black)) +
  geom_qq() +
  geom_qq_line() +
  labs(title = "Q-Q Plot Prop Black",
        x = "Theoretical Quantiles",
        y = "Sample Quantiles") +
  theme_minimal()+theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank())

p4_lion <- ggplot(lion_data, aes(sample = age_years)) +
  geom_qq() +
  geom_qq_line() +
  labs(title = "Q-Q Plot for age",
        x = "Theoretical Quantiles",
        y = "Sample Quantiles") +
  theme_minimal()

# Combine the plots
(p1_lion / p2_lion) | (p3_lion / p4_lion)

```



Lecture 9: Correlation Analysis

To understand the amount of variation explained, you can square the non parametric Spearman's rho value.

For your value of 0.74485:

$$\rho^2 = 0.74485^2 = 0.5548$$

This means approximately 55.48% of the variance in ranks of one variable can be explained by the ranks of the other variable. This is similar to how R^2 works in linear regression, but specifically for ranked data.

Spearman's rank correlation rho

```
data: lion_data$proportion_black and lion_data$age_years
S = 1392.1, p-value = 1.013e-06
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.7448561
```

Lecture 9: Correlation Analysis

Correlation: Important Considerations

The correlation coefficient depends on the range

- Restricting range of values can reduce the correlation coefficient
- Comparing correlations between studies requires similar ranges of values

Measurement error affects correlation

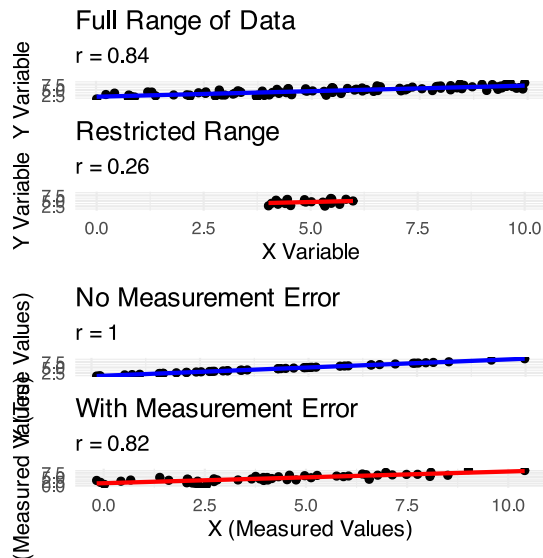
- Measurement error in X or Y tends to weaken observed correlation
- This bias is called **attenuation**
- True correlation typically stronger than observed correlation

Correlation vs. Causation

- Correlation does not imply causation
- Three possible explanations for correlation:
 1. X causes Y
 2. Y causes X
 3. Z (a third variable) causes both X and Y

Correlation significance test

- $H_0: \rho = 0$ (no correlation in population)
- $H_1: \rho \neq 0$ (correlation exists in population)
- **Test statistic: $t = r / SE(r)$ with $df = n-2$**



Lecture 9: Linear Regression

Simple Linear Regression Model

Simple linear regression models the relationship between a response variable (Y) and a predictor variable (X).

The **population** regression model

$$Y = \alpha + \beta X + \varepsilon$$

Where:

- Y is the response variable
- X is the predictor variable
- α (alpha) is the intercept (value of Y when $X=0$)
- β (beta) is the slope (change in Y per unit change in X)
- ε (epsilon) is the error term (random deviation from the line)

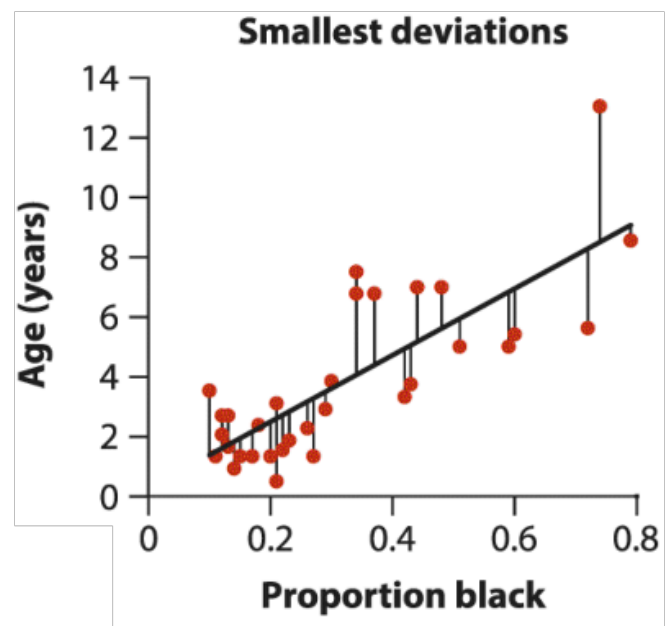
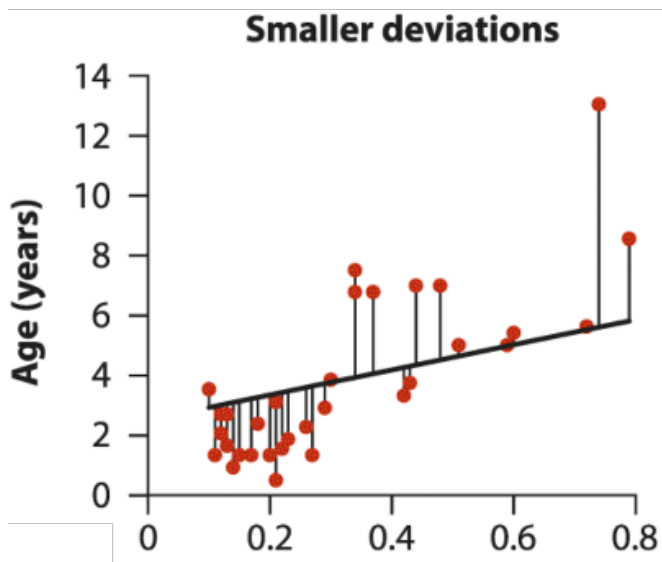
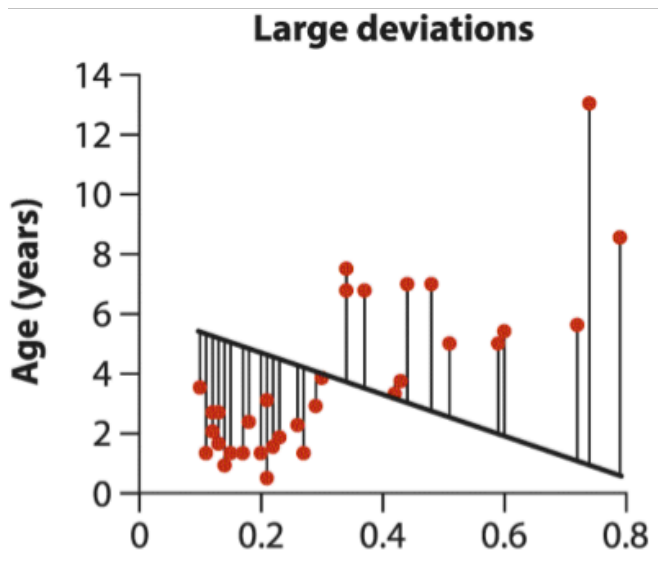
The **sample** regression equation is:

$$\hat{Y} = a + bX$$

Where:

- $\hat{Y}$ is the predicted value of Y
- a is the estimate of α (intercept)
- b is the estimate of β (slope)

Method of Least Squares: The line is chosen to minimize the sum of squared vertical distances (residuals) between observed and predicted Y values.



Lecture 9: Linear Regression

Simple Linear Regression Model

Simple linear regression models the relationship between a response variable (Y) and a predictor variable (X).

The **population** regression model is:

$$Y = \alpha + \beta X + \varepsilon$$

Where:

- Y is the response variable
- X is the predictor variable
- α (alpha) is the intercept (value of Y when X=0)
- β (beta) is the slope (change in Y per unit change in X)
- ε (epsilon) is the error term (random deviation from the line)

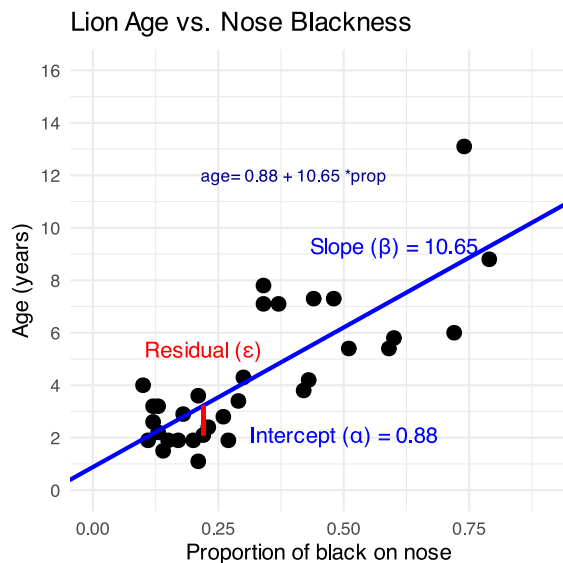
The **sample** regression equation is:

$$\hat{Y} = a + bX$$

Where:

- $\hat{Y}$ is the predicted value of Y
- a is the estimate of α (intercept)
- b is the estimate of β (slope)

Method of Least Squares: The line is chosen to minimize the sum of squared vertical distances (residuals) between observed and predicted Y values.

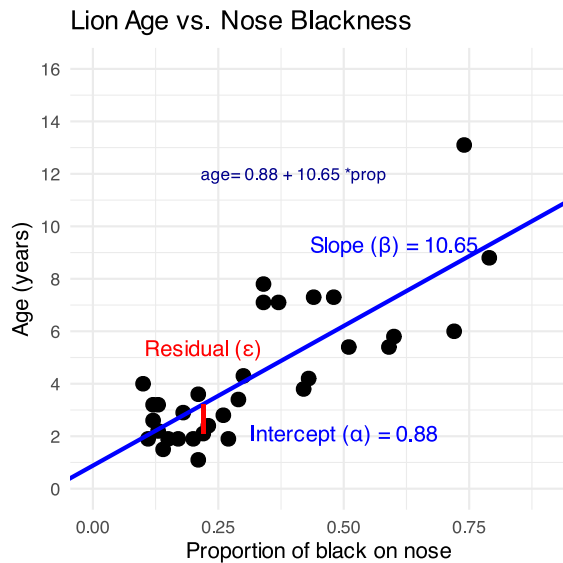


Lecture 9: Linear Regression

From Example 17.1 in the textbook the regression line for the lion data is:

$$\text{age} = 0.88 + 10.65 \times \text{proportion}_{\text{black}}$$

This means: - When a lion has no black on its nose (proportion = 0), its predicted age is 0.88 years - For each 0.1 increase in the proportion of black, age increases by 1.065 years - The slope (10.65) indicates that lions with more black on their noses tend to be older



Lecture 9: Linear Regression

Simple Linear Regression Model

- male lions develop more black pigmentation on their noses as they age.
- can be used to estimate the age of lions in the field.

Call:

```
lm(formula = age_years ~ proportion_black, data = lion_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.5449	-1.1117	-0.5285	0.9635	4.3421

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.8790	0.5688	1.545	0.133
proportion_black	10.6471	1.5095	7.053	7.68e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.669 on 30 degrees of freedom

Multiple R-squared: 0.6238, Adjusted R-squared: 0.6113

F-statistic: 49.75 on 1 and 30 DF, p-value: 7.677e-08

Lecture 9: Linear Regression

Simple Linear Regression Model

The calculation for slope (b) is:

$$b = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$$

Given: - $\bar{X} = 0.3222$ - $\bar{Y} = 4.3094$ - $\sum_i (X_i - \bar{X})^2 = 1.2221$ - $\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 13.0123$

$b = 13.0123 / 1.2221 = 10.647$

Intercept (a): $a = \bar{Y} - b\bar{X} = 4.3094 - 10.647(0.3222) = 0.879$

Making predictions:

To predict the age of a lion with 0.50 proportion of black on its nose:

$$\hat{Y} = 0.88 + 10.65(0.50) = 6.2 \text{ years}$$

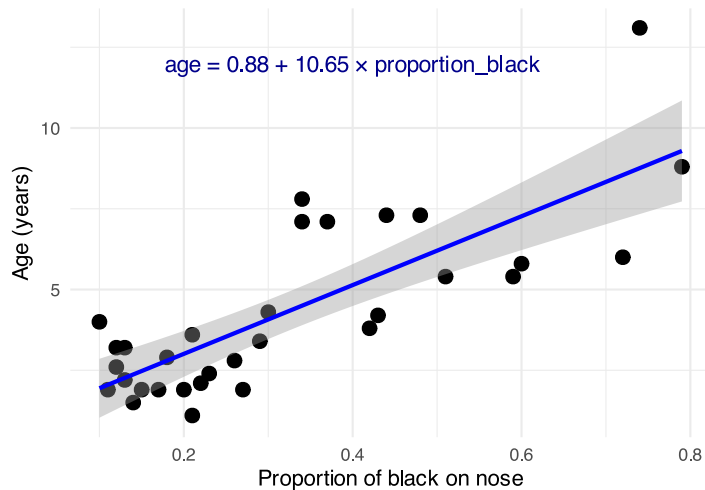
Confidence intervals vs. Prediction intervals:

- **Confidence interval:** Range for the mean age of all lions with 0.50 black
- **Prediction interval:** Range for an individual lion with 0.50 black

Both intervals are narrowest near $\bar{X}$ and widen as X moves away from the mean.

Lion Age vs. Nose Blackness

Using nose pigmentation to estimate age



Lecture 9: Linear Regression

Example Prairie Home Companion

- Does biodiversity affect ecosystem stability?
- Tilman et al. (2006) investigated using experimental plots varying plant species

```
# A tibble: 6 × 2
  species_number log_stability
      <dbl>         <dbl>
1           1         0.763
2           1         1.45
3           1         1.51
4           1         0.747
5           1         0.983
6           1         1.12
```

Call:

```
lm(formula = log_stability ~ species_number, data = prairie_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.82774	-0.25344	-0.00426	0.27498	0.75240

Coefficients:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.252629   0.041023  30.535 < 2e-16 ***
species_number 0.025984   0.004926   5.275 4.28e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3433 on 159 degrees of freedom
Multiple R-squared:  0.149, Adjusted R-squared:  0.1436
F-statistic: 27.83 on 1 and 159 DF, p-value: 4.276e-07

```

```
[1] "rsquared is: 0.148953385305455"
```

Analysis of Variance Table

```

Response: log_stability
              Df Sum Sq Mean Sq F value    Pr(>F)
species_number  1  3.2792   3.2792  27.829 4.276e-07 ***
Residuals     159 18.7358   0.1178
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Lecture 9: Linear Regression

The hypothesis test asks whether the slope equals zero:

- $H_0: \beta = 0$ (species number does not affect stability)
- $H_1: \beta \neq 0$ (species number does affect stability)

The test statistic is: $t = \frac{b - \beta_0}{SE_b}$

With $df = n - 2 = 161 - 2 = 159$

Interpretation:

The slope estimate is 0.033, indicating that log stability increases by 0.033 units for each additional plant species in the plot.

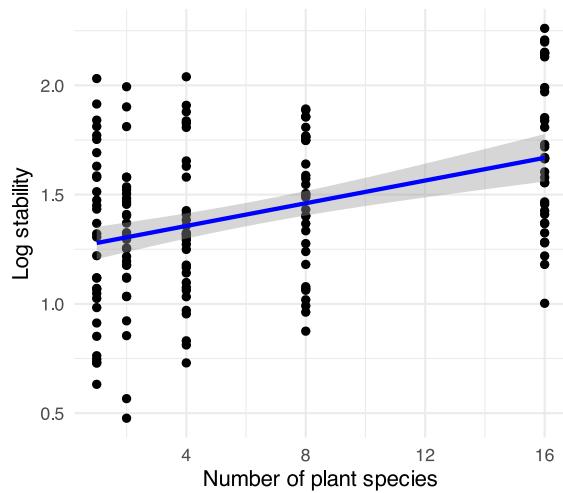
The p-value is very small ($2.73e-10$), providing strong evidence to reject the null hypothesis that species number has no effect on ecosystem stability.

$R^2 = 0.222$, meaning that approximately 22.2% of the variation in log stability is explained by the number of plant species.

This supports the biodiversity-stability hypothesis: more diverse plant communities have more stable biomass production over time.

Biodiversity and Ecosystem Stability

$R^2 = 0.149$



Lecture 9: Linear Regression

Testing Regression Assumptions

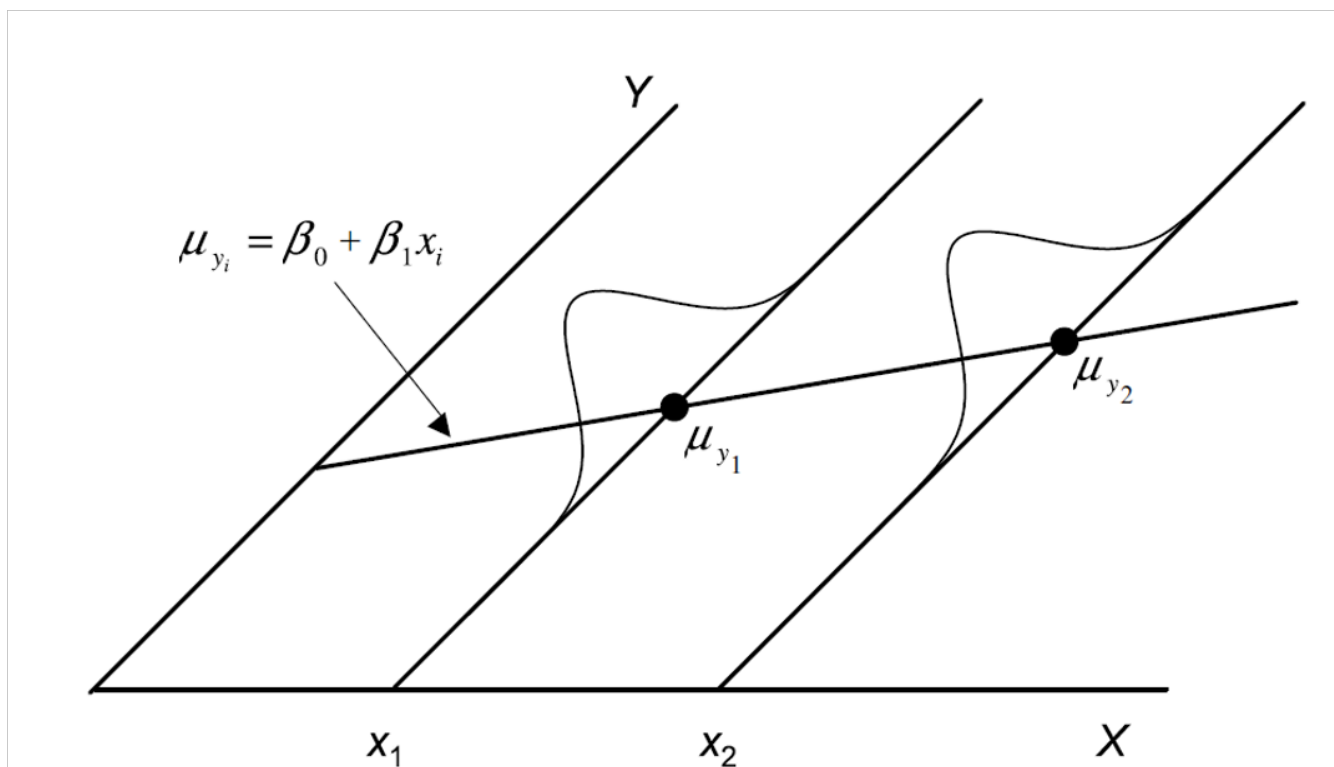
linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:

Assume that **error** e_i is $e_i = y_i - \hat{y}_i$

- normally distributed for each x_i
- has the same variance
- has a mean of 0 at each x_i



Lecture 9: Linear Regression

Testing Regression Assumptions

linear regression has four key assumptions:

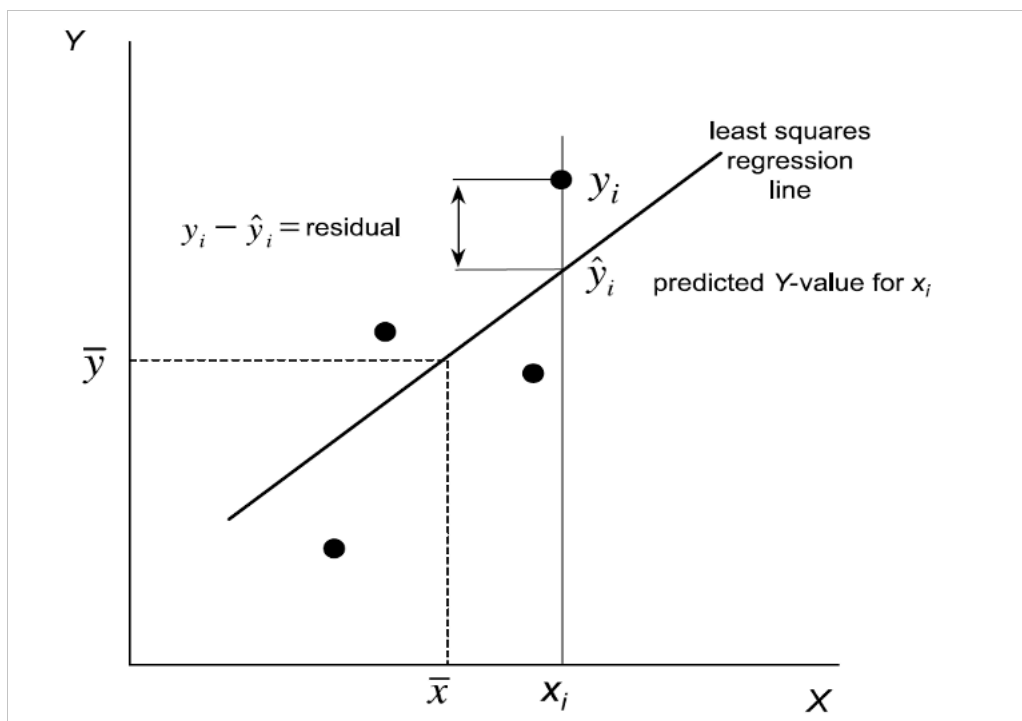
1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:

Assume that **error** ϵ_i is - estimated as the residuals: $e_i = y_i - \hat{y}_i$

- ordinary least square estimates a and b or slope and intercept to minimize the sum of the residuals squared or Mean Squared Error (MSE) as

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$



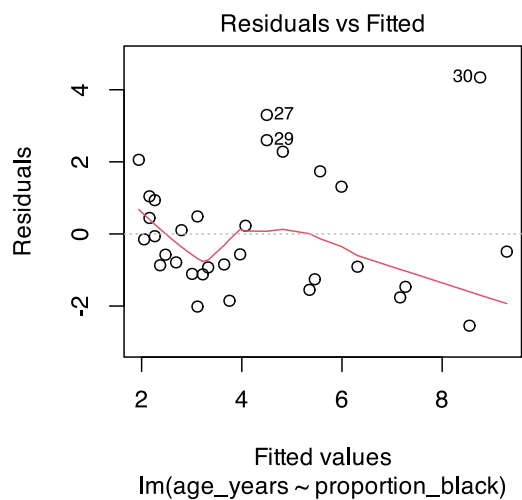
Lecture 9: Linear Regression

Testing Regression Assumptions

linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:



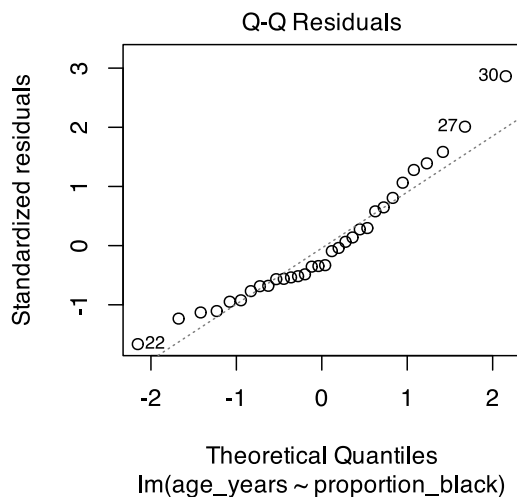
Lecture 9: Linear Regression

Testing Regression Assumptions

linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:



Lecture 9: Linear Regression

Testing Regression Assumptions

linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

Let's check these assumptions for the lion regression model:

Shapiro-Wilk normality test

```
data: residuals(lion_model)
W = 0.93879, p-value = 0.0692
```

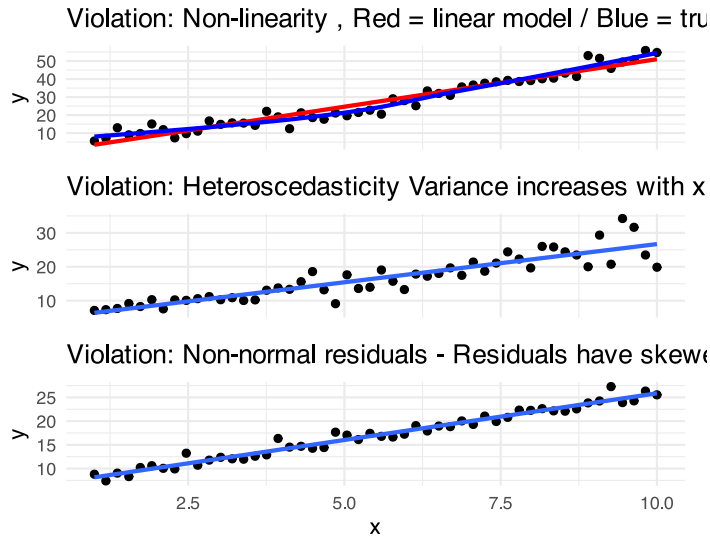
Lecture 9: Linear Regression

Simple Linear Regression Model

linear regression has four key assumptions:

1. **Linearity:** The relationship between X and Y is linear
2. **Independence:** Observations are independent
3. **Homoscedasticity:** Equal variance across all values of X
4. **Normality:** Residuals are normally distributed

If assumptions are violated: 1. Transform the data (Section 17.6) 2. Use weighted least squares for heteroscedasticity 3. Consider non-linear models (Section 17.8)



Lecture 9: Linear Regression - estimates of error and significance

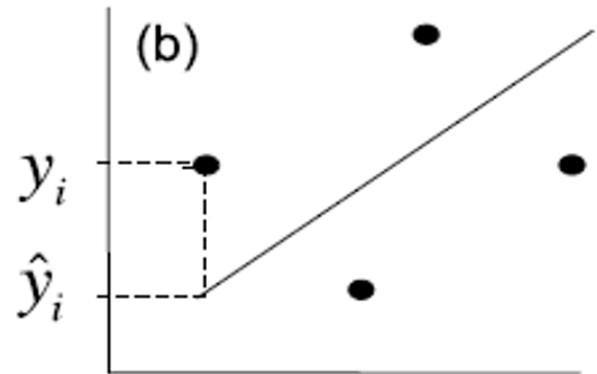
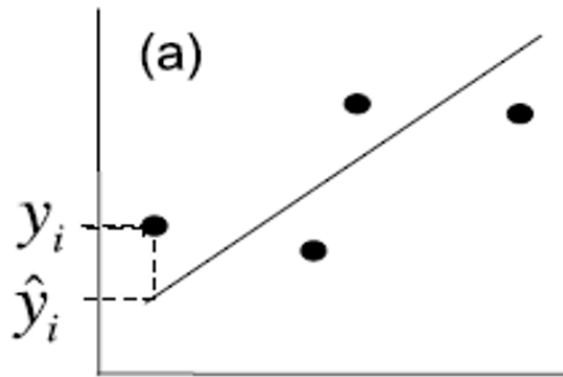
- Estimates of standard error and confidence intervals for slope and intercept to determine confidence bands
- the 95% confidence band will contain the true population line 95/100 under repeated sampling
- this is usually done in R

Parameter	OLS estimate	Standard error
β_1	$b_1 = \frac{\sum_{i=1}^n [(x_i - \bar{x})(y_i - \bar{y})]}{\sum_{i=1}^n (x_i - \bar{x})^2}$	$s_{b_1} = \sqrt{\frac{MS_{\text{Residual}}}{\sum_{i=1}^n (x_i - \bar{x})^2}}$
β_0	$b_0 = \bar{y} - b_1 \bar{x}$	$s_{b_0} = \sqrt{MS_{\text{Residual}} \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]}$
ε_i	$e_i = y_i - \hat{y}_i$	$\sqrt{MS_{\text{Residual}}} \text{ (approx.)}$

Lecture 9: Linear Regression - estimates of error and significance

In addition to getting estimates of population parameters (β_0 , β_1), want to test hypotheses about them

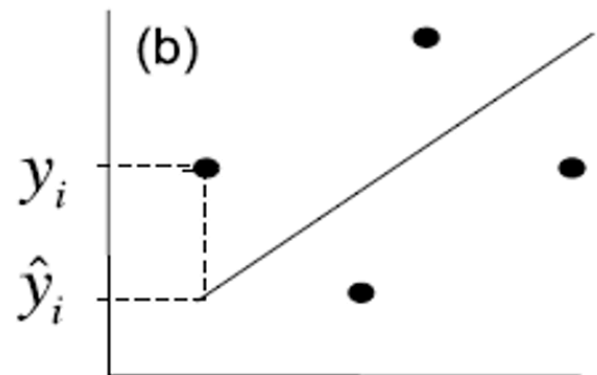
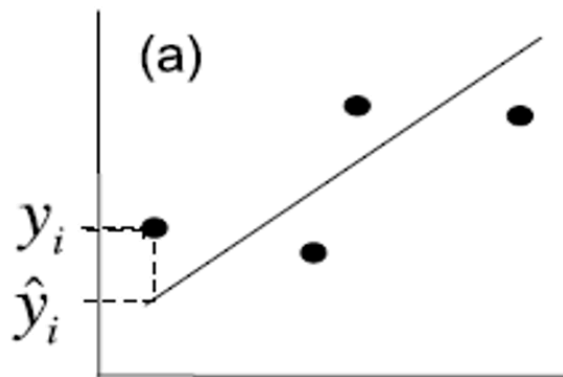
- This is accomplished by analysis of variance
- Partition variance in Y: due to variation in X, due to other things (error)



Lecture 9: Linear Regression - estimates of variance

Total variation in Y is “partitioned” into 3 components:

- $SS_{regression}$: variation explained by regression
 - difference between predicted values ($\hat{y}_i$) and mean y ($\bar{y}$)
 - $dfs = 1$ for simple linear (parameters-1)
- $SS_{residual}$: variation not explained by regression
 - difference between observed (y_i) and predicted ($\hat{y}_i$) values
 - $dfs = n - 2$
- SS_{total} : total variation
 - sum of squared deviations of each observation (y_i) from mean ($\bar{y}$)
 - $dfs = n - 1$



Lecture 9: Linear Regression - estimates of variance

Total variation in Y is “partitioned” into 3 components:

- $SS_{regression}$: variation explained by regression
 - difference between predicted values ($\hat{y}_i$) and mean y ($\bar{y}$)
 - $dfs = 1$ for simple linear (parameters-1)
- $SS_{residual}$: variation not explained by regression
 - difference between observed (y_i) and predicted ($\hat{y}_i$) values
 - $dfs = n - 2$
- SS_{total} : total variation
 - sum of squared deviations of each observation (y_i) from mean ($\bar{y}$)

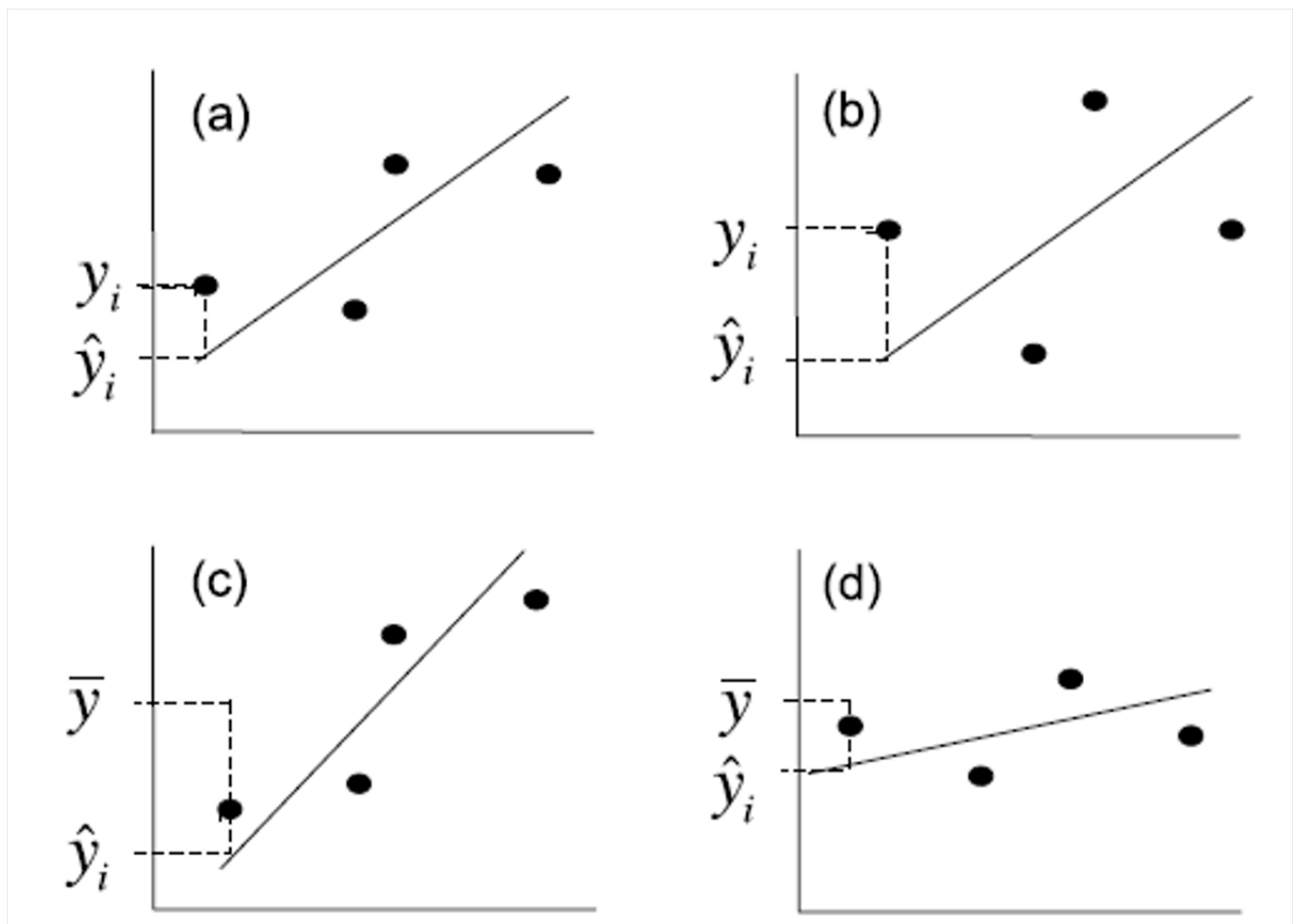
▸ $dfs = n-1$

Source of variation	SS	df	MS	Expected mean square
Regression	$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$	1	$\frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{1}$	$\sigma_\varepsilon^2 + \beta_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$
Residual	$\sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n-2$	$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}$	σ_ε^2
Total	$\sum_{i=1}^n (y_i - \bar{y})^2$	$n-1$		

Lecture 9: Linear Regression - estimates of variance

Total variation in Y is “partitioned” into 3 components:

- $SS_{regression}$: variation explained by regression
 - GREATER IN C than D
- $SS_{residual}$: variation not explained by regression
 - GREATER IN B THAN A
- SS_{total} : total variation



Lecture 9: Linear Regression - estimates of variance

Sums of Squares and degrees of freedom are:

$$SS_{regression} + SS_{residual} = SS_{total}$$

$$df_{regression} + df_{residual} = df_{total}$$

- Sums of Squares depends on n
- We need a different estimate of variance

Source of variation	SS	df	MS	Expected mean square
Regression	$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$	1	$\frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{1}$	$\sigma_\varepsilon^2 + \beta_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$
Residual	$\sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n - 2$	$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2}$	σ_ε^2
Total	$\sum_{i=1}^n (y_i - \bar{y})^2$	$n - 1$		

Lecture 9: Linear Regression - estimates of variance

Sums of Squares converted to Mean Squares

- Sums of Squares divided by degrees of freedom - does not depend on n
- $MS_{residual}$: estimate population variation
- $MS_{regression}$: estimate pop variation and variation due to X-Y relationship
- Mean Squares are not additive

Source of variation	SS	df	MS	Expected mean square
Regression	$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$	1	$\frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{1}$	$\sigma_\varepsilon^2 + \beta_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$
Residual	$\sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n - 2$	$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2}$	σ_ε^2
Total	$\sum_{i=1}^n (y_i - \bar{y})^2$	$n - 1$		